SEHAJ SINGH

AI/ML ARCHITECT & EDUCATOR · GENERATIVE AI & LEARNING SYSTEMS

Diamond Bar, CA · (+1) 840-230-0561 · sehajroop.singh@hotmail.com · linkedin.com/in/sehajroop-singh · github.com/sehajroopsingh

PROFILE

AI/ML architect and educator with end-to-end experience designing, deploying, and governing production Generative AI systems — agentic LLM workflows, RAG pipelines, evaluation frameworks, and production guardrails. A product of the California Community Colleges (A.A., Santa Monica College), now teaching at Mt. San Antonio College and Cal Poly Pomona. Founder of MemSurf, an AI-native learning platform built around multilingual access and affordability; proposed and led a GenAI feasibility and governance assessment inside Johnson & Johnson MedTech, an FDA-regulated Fortune-50 enterprise. Combines hands-on LLM engineering with AI governance, translating complex technical work for students, faculty, administrators, and executive leadership.

EXPERIENCE

Adjunct Faculty — California State Polytechnic University, Pomona **Aug 2026 – Present**
Computer Science Department *Pomona, CA*

- **Undergraduate instruction:** teaching Fundamentals of Data Science — a foundation course for the university's Artificial Intelligence minor — and Introduction to Web Science & Technology: lectures, guided problem-solving, grading, and structured student feedback.
- **AI-era course design:** structuring assignments in both courses so students build the AI fluency their workplaces will expect — using AI where it scaffolds learning, working unassisted where it would substitute for it.

Adjunct Faculty — Mt. San Antonio College **Summer 2026**
High School Referral (HSR) Summer Program *Walnut, CA*

- **Teaching & mentoring:** taught 130+ high school students across 2 course sections in the High School Referral (HSR) Summer Program, an early college-access pathway — owning instruction end-to-end: breaking complex concepts into structured explanations, guiding hands-on problem-solving, managing pacing, assessment, and grading, and adapting delivery to widely varied levels of preparation — while supervising a teaching assistant and coordinating grading standards, assignment quality, and assessment across both sections.

Founder & AI/ML Systems Architect — MemSurf **2025 – Present**
AI-native learning platform (web, iOS, Android) *Los Angeles, CA*

- **End-to-end GenAI architecture:** faced with turning arbitrary user-uploaded content into reliable study material, designed and shipped the full platform solo — Python/Django/DRF services, Celery distributed task queues, PostgreSQL + pgvector semantic search, WebSockets, and native SwiftUI (iOS) and Kotlin (Android) clients, with models deployed and served in production via Google Vertex AI. Now live with 100+ active users and thousands of AI-generated lessons, quizzes, and content items processed.
- **Agentic LLM workflows as services:** single-prompt LLM calls couldn't reliably handle multi-step jobs (parse content → generate lessons → build quizzes → personalize review), so built multi-step agentic workflows (LangChain, tool-calling, RAG grounded in the user's own content) exposed as asynchronous micro-service workers behind clean API contracts — then parallelized them across Celery multi-threaded workers, cutting end-to-end pipeline completion time by ~98% (50x faster) versus sequential calling while preserving per-call reliability.
- **Production guardrails & LLM evaluation:** raw LLM calls produced malformed typed outputs and inconsistent quality at unpredictable cost, so engineered production guardrails — structured-output validation against typed schemas and business rules, automatic retry-and-repair, model benchmarking against quality/latency/cost budgets, and an LLM-as-judge grading pipeline — raising end-to-end generation success from near-zero at the start to 90% first-pass and 99%+ after automated retries, and cutting cost per generation by ~80% (5x+) through task decomposition: routing short, scoped calls to cheaper models with prompt designs that preserved output quality — cost engineering with a purpose: keeping the platform free, ad-supported, and multilingual for learners who cannot pay for subscriptions, while meeting U.S. state privacy requirements and EU GDPR.
- **Applied ML personalization:** one-size-fits-all review wastes learner effort on already-mastered material while under-drilling weak spots, so built content-aware memory models, adaptive-difficulty logic, and knowledge-graph-inspired learning paths that recompose each user's review queue in real time — concentrating practice on their weakest concepts instead of uniform repetition, and powering the spaced-repetition loop at the core of the product.

Software Engineer (Contract) — Johnson & Johnson (MedTech) **Apr 2024 – Sep 2024**
Abiomed / J&J MedTech post-acquisition integration — regulated medical-device enterprise *Irvine, CA*

- **Enterprise systems integration (post-acquisition):** drove the systems analysis for onboarding the newly acquired Abiomed Heart Recovery business onto J&J's enterprise Service & Repair platform — mapping device records, service and repair workflows, integration middleware, vendor data feeds, and the business and validation rules governing them, so the acquired company could inherit the capabilities of a Fortune-50 enterprise stack. Owned the issue flow through onboarding and validation: triaging integration failures as they surfaced, escalating blockers to the right leaders before they cost schedule, keeping every stakeholder current, and driving fixes to closure.
- **Generative AI proof of concept & governance assessment:** identified employee training-content comprehension as a high-cost workflow without clear ownership — across J&J's ~138,000-person workforce, a 20-hour annual saving per employee at loaded professional cost (~$96/hour) compounds to over $260M — and proposed the initiative to address it. Architected the proof of concept for the enterprise environment in which it would operate: built directly over the existing training-content data store, designed within J&J's infrastructure, data landscape, and security and compliance boundaries, introducing generative AI capability without re-engineering legacy systems. Benchmarked baseline training-content review time against AI-assisted retrieval, then presented the business and technical case — measured baseline, projected time savings per employee, and implementation and data-privacy risk in an FDA-regulated environment — to senior leadership, securing approval for initial development and delivering a working prototype operating against the production training-content store.

Software Engineer (Internship) — FotonVR **Feb 2024 – Mar 2024**
EdTech / VR learning content *Remote*

- **Syllabus-coverage Graph-RAG engine:** FotonVR needed to know whether its VR lesson library actually covered any given school's curriculum; built an LLM-agent pipeline that decomposed syllabi into discrete learning outcomes, matched each outcome against the existing VR content library via natural-language-to-Neo4j graph queries and graph-based RAG, and flagged uncovered outcomes as a prioritized new-content roadmap — working directly with the CTO on retrieval strategy and evaluation.

Business Analyst (Internship) — Think AI **May 2023 – Dec 2023**
Business intelligence over enterprise customer data *Remote*

- **Requirements-to-metrics translation:** partnered with 8 business stakeholders to convert loosely framed requests into measurable KPIs — defining primary success metrics alongside counter-metrics to prevent misleading wins — and built and maintained 15+ Power BI reports and the underlying SQL analysis, adopted by 4 teams for ongoing decision-making.

Executive Board Member / Interim Executive Director — UCLA Cooperative Housing Assoc. **2018 – 2020**
$3M/yr nonprofit serving 500+ student residents *Los Angeles, CA*

- **Organizational leadership:** held budget-allocation authority for a $3M/year operation, setting funding priorities across operations and multi-year projects; led the board's formal review of executive performance, assuming Interim Executive Director duties for organizational continuity and partnering with lawyers and accountants throughout.
- **COVID-19 response:** designed and implemented the cooperative's pandemic response policy for a high-density, high-risk residential facility, translating evolving UCLA, state, and federal requirements into operational policy across multiple departments. Assessed organizational objectives and specific challenges facing cleaning, security, kitchen, retail, and other functions; met twice weekly with department leads, managers, and staff, building strategic alignment, developing workable solutions, and iteratively adjusting policies. The response resulted in zero infections over 10 months, with elements adopted by peer student cooperatives in Berkeley and Santa Barbara within the North American Students of Cooperation (NASCO) network.
- **Community governance (Membership Committee, 2018–19):** mediated member disputes and made case-by-case policy decisions in a democratically run housing cooperative — balancing individual circumstances against community rules and organizational constraints, while leading external outreach.

Care Extender, Emergency Dept. & Oncology — UCLA Ronald Reagan Medical Center **2017 – 2018**
Care Extender Internship Program — UCLA Health *Los Angeles, CA*

- **Patient care team member:** completed the year-long Care Extender clinical internship as a trained member of patient care teams in the Emergency Department and Oncology — assisting nurses with hands-on patient care, transfers, and transport, supporting patients and families through high-stress moments, working under HIPAA and infection-control protocols.

English as a Second Language Instructor — Culturalía México **Aug 2015 – Jun 2016**
English instruction in underserved communities *Comitán de Domínguez, Chiapas, Mexico*

- **Equity-centered instruction:** taught English full-time to 100+ students — designing lesson plans for learners whose culture, resources, and preparation differed from my own, assessing progress across the term, and preparing students for goals including TOEFL exams for study abroad.

EDUCATION

M.S. Computer Science — Georgia Institute of Technology **2023 – 2025**
Relevant Coursework: Machine Learning · Computer Vision · Artificial Intelligence for Robotics · Educational Technology · Advanced Operating Systems · Human Computer Interaction · Introduction to Information Security

B.S. Statistics & Data Science — UC Santa Barbara **2020 – 2023**
Relevant Coursework: Data Structures & Algorithms · Operating Systems · Computer Security · Network Security · General Chemistry · Organic Chemistry · Probability · Mathematical Statistics · Regression Analysis · Experimental Design. *Study abroad:* University of Edinburgh (Artificial Intelligence, Computer Security)

A.A. General Science — Santa Monica College **2016 – 2020**
Additional California community college coursework: Mt. San Antonio College · Berkeley City College

TECHNICAL SKILLS

Generative AI & LLM Engineering: LLM APIs (OpenAI, Anthropic, Gemini), Google Vertex AI (production model deployment), Azure AI Foundry (model deployment), LangChain, RAG, agent/tool-calling workflows, prompt engineering, semantic & vector search (pgvector, Neo4j), embeddings, LLM evaluation, adaptive learning algorithms

AI/ML Frameworks: Python, Scikit-learn, TensorFlow/Keras, PyTorch, NLP (applied), model evaluation & experimentation

AI Governance & Guardrails: AI feasibility & risk assessment, compliance-aware architecture in regulated (FDA MedTech) environments, FERPA-aware data handling, guardrail design (schema & business-rule output validation, grounded/RAG generation, retry-fallback logic, cost & latency budgets), LLM evaluation frameworks (LLM-as-judge), human-in-the-loop review design, enterprise adoption roadmapping

Backend & Data Architecture: Django/DRF, Celery, PostgreSQL, REST APIs, WebSockets, asynchronous processing, data pipelines, API design, microservices, enterprise system integration

Cloud & Infrastructure: Google Cloud (Vertex AI), Azure (AI Foundry), AWS, Docker, distributed task queues, background workers, observability, CI/CD

Impact Measurement & Analytics: KPI and metrics frameworks — defining success metrics, counter-metrics, and evaluation baselines for programs and products; multi-stage A/B testing and experimentation for causal accuracy; SQL, Power BI & DAX (Microsoft Fabric BI layer), Amplitude product analytics, event tracking, dashboarding, stakeholder reporting

Domain Context: Regulated healthcare / MedTech software, education technology, learning analytics, California Community Colleges & CSU instruction